AI in Context: The Rise of Learning

chris.wiggins@columbia.edu

Fall 2025 - Lecture 4 of 4

Today's Journey: From Rules to Learning

The Big Question

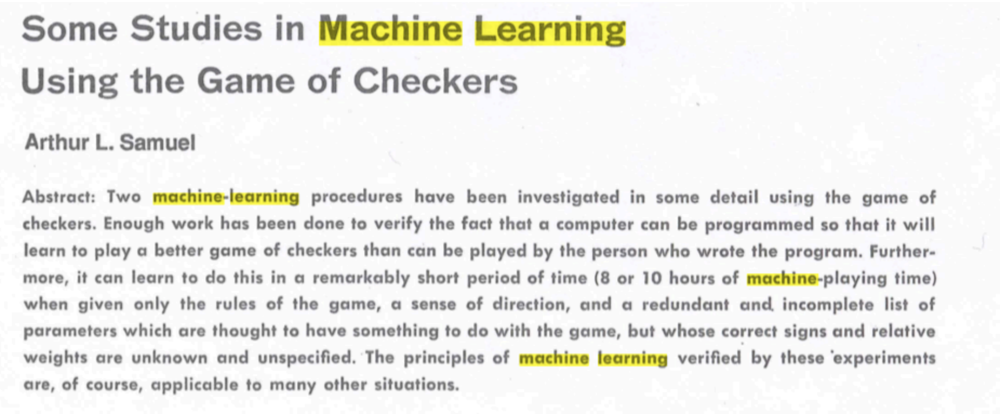
How did we get from "just program it" to "just train it"?

Our Path

1. **The Birth of Machine Learning** (1959-1980s)
2. **The Long Winter** (1980s-2000s)
3. **The Great Awakening** (2012-2022)
4. **What We Gained, What We Lost**

Part I: The Birth of Machine Learning (1959-1980s)

1959: The Term "Machine Learning" is Born



Some Studies in Machine Learning Using the Game of Checkers

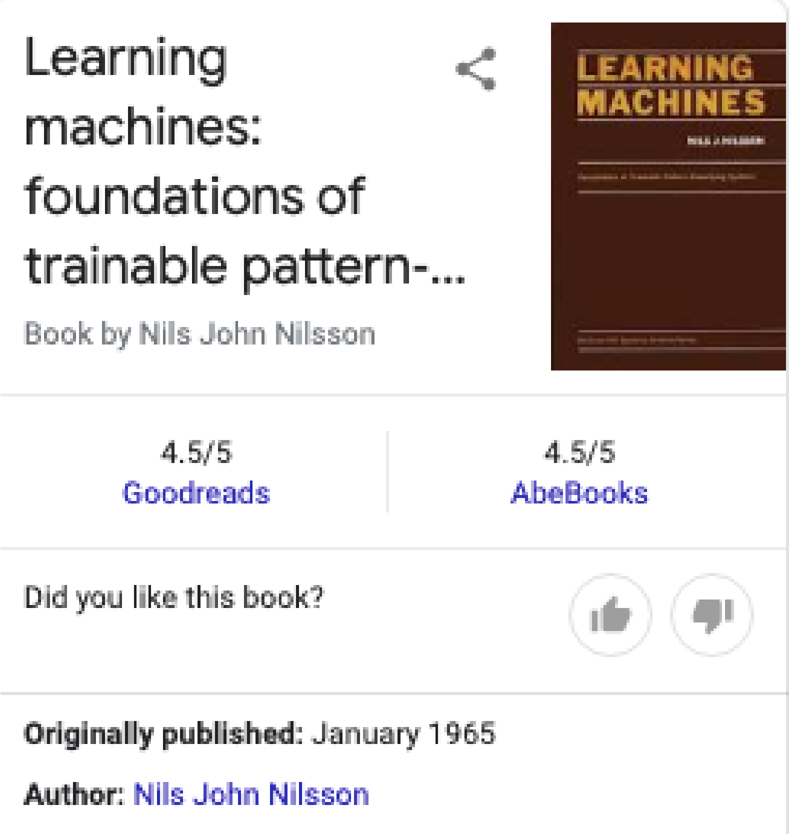
Arthur L. Samuel

Abstract: Two machine-learning procedures have been investigated in some detail using the game of checkers. Enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program. Furthermore, it can learn to do this in a remarkably short period of time (8 or 10 hours of machine-playing time) when given only the rules of the game, a sense of direction, and a redundant and incomplete list of parameters which are thought to have something to do with the game, but whose correct signs and relative weights are unknown and unspecified. The principles of machine learning verified by these experiments are, of course, applicable to many other situations.

Arthur Samuel at IBM: "Some studies in machine learning using the game of checkers"

- **Key insight**: Machines can improve through experience
- **Method**: Self-play and evaluation
- **Demo**: 1956 TV appearance with IBM 701

But ML Didn't Take Off Like AI Did



Why? Neural networks were embraced by **pattern recognition** (EE), not AI (CS)

- **AI community**: Focused on symbolic logic and rules
- **Pattern recognition**: Statistical approaches, signal processing

The Pattern Recognition Community (1950s-1970s)

Key Figures & Ideas

1955: Oliver Selfridge - "Pattern recognition and modern computers"

1968: George Nagy (Rosenblatt's student) - Train/test split

1972: Jerry Friedman - "It was called pattern recognition then; it's called machine learning now"

Core Methods

- Statistical signal detection
- Discriminant analysis
- Nearest neighbor
- Decision trees

The Statistical Foundation

> *"Statistical Signal Detection theory was adopted in the pattern recognition community at a very early stage. Chow (1957) proposed using optimal detection theory, and this led to a proposal by Highleyman to approximate the risk by its sample average. This transition from population risk to 'empirical' risk gave rise to what we know today as machine learning."*

-- *mlstory.org*

Military Applications Drive Development

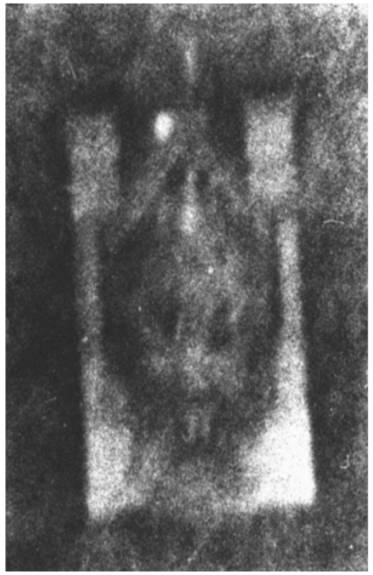


Figure 4.13: A typical tank image. (Photograph courtesy of Thomas Harley.)

DARPA-funded projects:

- **Shakey the robot** (1966-1972): Autonomous sentry
- **SRI**: Computer vision for military
- **Pattern recognition**: Target identification, signal processing

Part II: The Long Winter (1980s-2000s)

The 1980s: Machine Learning Emerges as a Field

1983: The Journal of Machine Learning

Pat Langley's vision:

> *"Historically, researchers have taken two approaches to machine learning. Numerical methods such as discriminant analysis have proven quite useful in perceptual domains, and have become associated with the paradigm known as Pattern Recognition. In contrast, Artificial Intelligence researchers have concentrated on symbolic learning methods"*

Missing: Statistics (the elephant in the room)

The Symbolic vs. Statistical Divide

Symbolic ML (AI Community)

- **Goal**: Build intelligent systems
- **Methods**: Expert systems, rule learning
- **Focus**: Knowledge representation
- **Problem**: Knowledge acquisition bottleneck

Statistical ML (Pattern Recognition)

- **Goal**: Make accurate predictions
- **Methods**: Neural networks, statistics
- **Focus**: Performance on benchmarks
- **Problem**: "Black box" methods

The Connectionist Revival (1980s)

Key Developments

- **Backpropagation** (1986): Rumelhart, Hinton, Williams
- **Parallel Distributed Processing** (PDP) group
- **Connectionism**: Neural networks as models of cognition

The Problem

- **Limited data**: Not enough examples
- **Limited compute**: Not enough processing power
- **Limited theory**: Why did it work?

The Statistical Turn (1990s)

Michael Jordan's Perspective (2009)

> *"Throughout the eighties and nineties, it was striking how many times people working within the 'ML community' realized that their ideas had had a lengthy pre-history in statistics."*

Key insight: ML was rediscovering statistical methods

- Decision trees, logistic regression, PCA
- Cross-validation, bootstrap, EM algorithm
- Bayesian inference, graphical models

The Field Splits (1990s-2000s)

Pat Langley's Observations (2011)

> *"In 1986, when we launched the journal, machine learning was still viewed as a branch of artificial intelligence. By 2000, many researchers committed to machine learning treated it as a separate field with few links to its parent discipline."*

The shift:

- **From**: Building intelligent systems
- **To**: Making accurate predictions
- **Result**: Performance over understanding

Part III: The Great Awakening (2012-2022)

2012: The ImageNet Moment

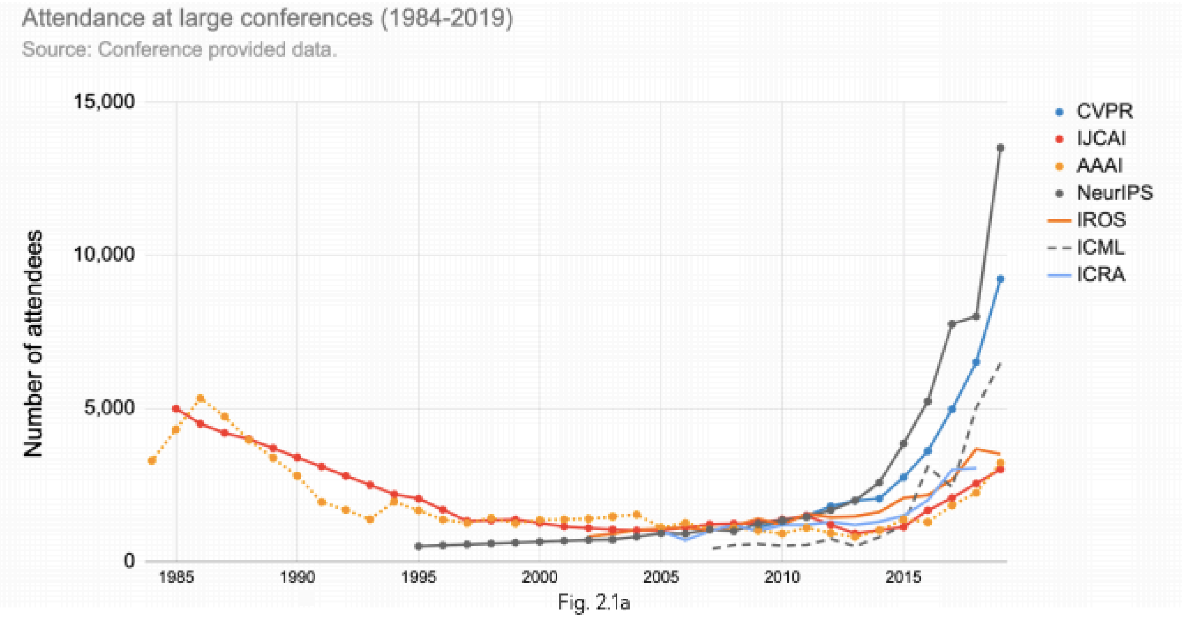


Fig. 2.1a

Note: IJCAI occurred every other year till 2014. The missing year between 1984 and 2014 are interpolated as the mean between the two known conference attendance dates to provide a comparative view across conferences.

AlexNet on ImageNet:

- **Error rate**: 15.3% (vs. 26.2% for second place)
- **Method**: Deep convolutional neural network
- **Hardware**: GPUs made training feasible

Why 2012? The Perfect Storm

The Three Ingredients

1. **Big Data**: Internet-scale datasets (ImageNet, web text)
2. **Big Compute**: GPUs, distributed computing
3. **Big Models**: Deep neural networks with millions of parameters

Hinton's insight: "There just wasn't enough computer power or enough data. People on our side kept saying, 'Yeah, but if I had a really big [computer], it would work.'"

The Deep Learning Revolution

Key Developments (2012-2017)

- **2012**: AlexNet (computer vision)
- **2014**: Word2Vec (natural language)
- **2015**: ResNet (deeper networks)
- **2017**: Transformer architecture
- **2018**: BERT (language understanding)

The pattern: Scale up everything - data, compute, model size

The Industry Transformation

Google's "Great AI Awakening" (2016)

> *"Google's decision to reorganize itself around A.I. was the first major manifestation of what has become an industry wide machine-learning delirium."*

The talent war:

- Seven-figure salaries for PhD students
- Corporate labs competing with universities
- Conference attendance quadrupled

Part IV: What We Gained, What We Lost

What We Gained

Capabilities

- **Computer vision**: Object recognition, image generation
- **Natural language**: Translation, generation, understanding
- **Prediction**: Better than humans on many tasks
- **Automation**: End-to-end learning systems

Methods

- **End-to-end learning**: Raw data → predictions
- **Transfer learning**: Pre-trained models
- **Self-supervision**: Learning from unlabeled data

What We Lost

Understanding

- **Interpretability**: Why do models work?
- **Causality**: Correlation vs. causation
- **Robustness**: Adversarial examples, distribution shift

Control

- **Transparency**: Black box decision making
- **Fairness**: Biased predictions
- **Safety**: Unintended consequences

The Two Cultures Problem

Leo Breiman's Warning (2001)

> *"In the mid-1980s two powerful **new** algorithms for fitting data became available: neural nets and decision trees. A new research community using these tools sprang up. Their goal was predictive accuracy."*

The divide:

- **Statistics**: Understanding and modeling
- **Machine Learning**: Prediction and performance

The Current State (2022)

Impressive Capabilities

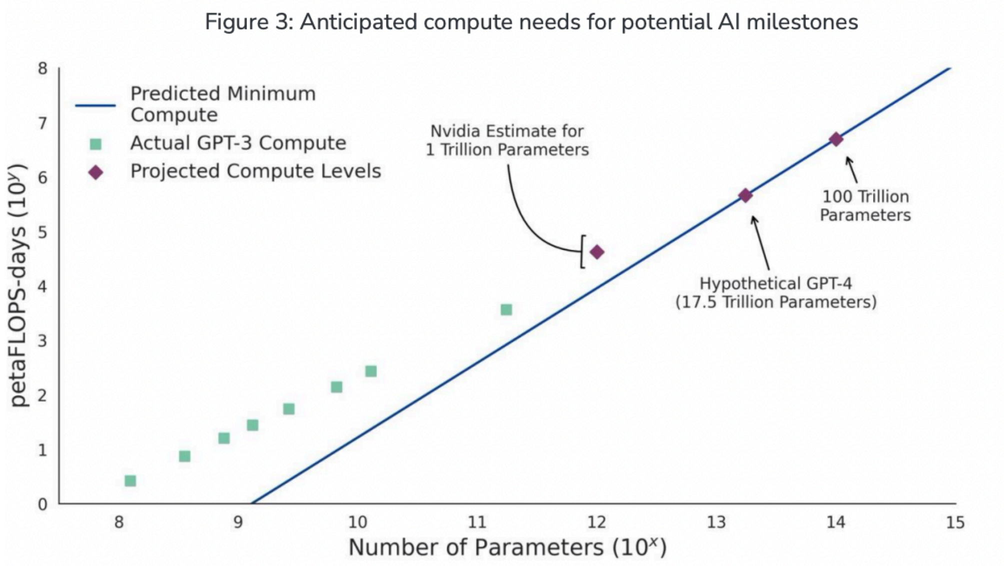
- **GPT-3**: Human-like text generation
- **Image generation**: DALL-E, Midjourney
- **Code generation**: GitHub Copilot
- **Multimodal**: Vision + language models

Persistent Problems

- **Hallucination**: Models make up facts
- **Bias**: Reflecting training data biases
- **Robustness**: Failing on edge cases
- **Interpretability**: Still largely black boxes

The Physical Limits

Scale Requirements



The trend: Exponential growth in model size, compute, and cost

The Environmental Cost

[Submitted on 5 Jun 2019]

Energy and Policy Considerations for Deep Learning in NLP

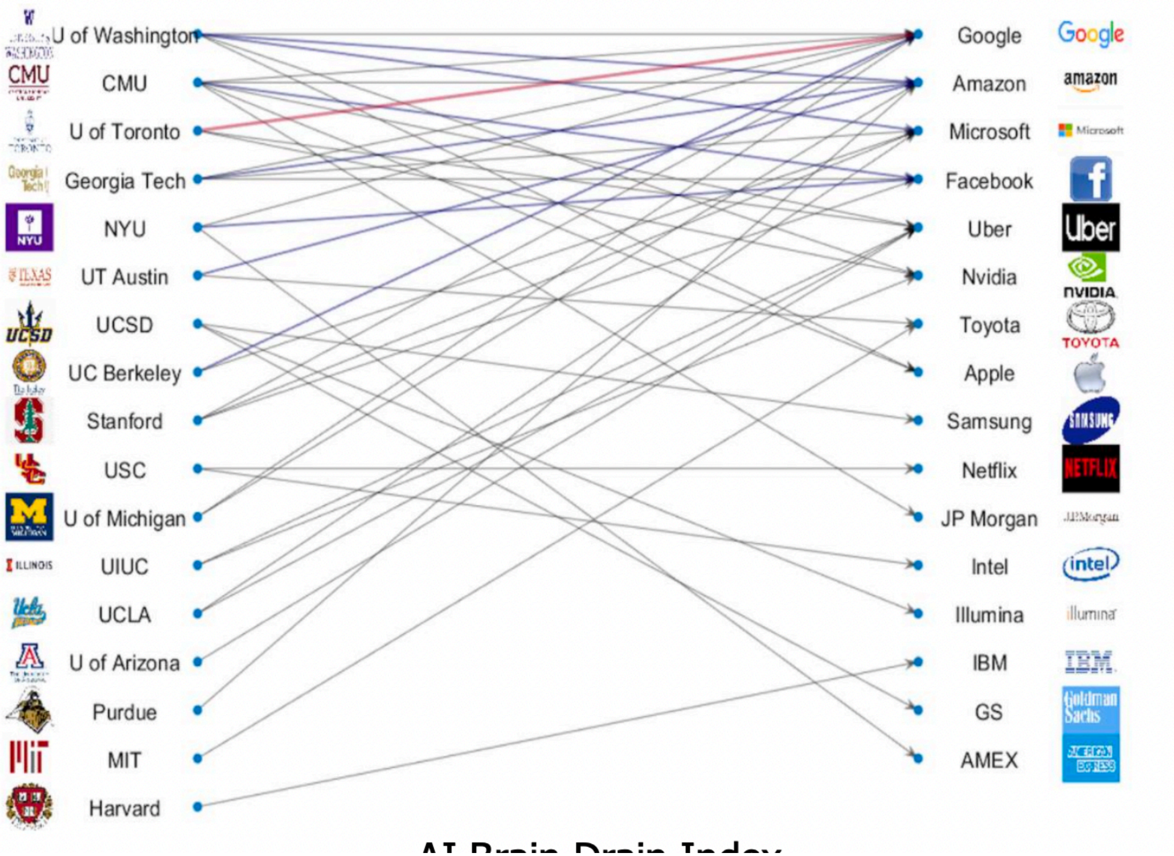
Emma Strubell, Ananya Ganesh, Andrew McCallum

Training large models:

- **GPT-3**: 1,300 MWh (equivalent to 120 US homes for a year)
- **Carbon footprint**: Growing concern about sustainability

Who's Doing AI?

The Brain Drain



Industry vs. Academia:

The Future: What's Next?

Current Challenges

1. **Alignment**: Making AI systems safe and beneficial
2. **Robustness**: Handling distribution shift and edge cases
3. **Efficiency**: Reducing compute and energy requirements
4. **Interpretability**: Understanding how models work

The Big Questions

- **AGI**: When (if ever) will we achieve general intelligence?
- **Control**: How do we maintain human agency?
- **Democratization**: Who gets to build and use these systems?

Looking Back: The Journey from Rules to Learning

The Arc of History

- **1950s-1980s**: Rules rule (GOFAI)
- **1980s-2000s**: The long winter (symbolic vs. statistical)
- **2012-2022**: The great awakening (deep learning)
- **2022+**: The reckoning (capabilities vs. control)

The Central Tension

Performance vs. Understanding: Can we have both?

Key Takeaways

What We've Learned

1. **Scale matters**: Big data + big compute + big models
2. **History repeats**: Old ideas become new again
3. **Context is everything**: Technology shapes society, society shapes technology
4. **Trade-offs are real**: Every gain comes with costs

The Path Forward

- **Balance**: Performance AND understanding
- **Responsibility**: Building systems that serve humanity
- **Wisdom**: Learning from history to avoid repeating mistakes

Questions?

Thank you for joining us for this journey through AI in Context!

Appendix: The Complete Timeline

1959-2012: The Long Road to Deep Learning

1959: Arthur Samuel coins "machine learning"

1960s: Pattern recognition community develops statistical methods

1970s: Neural networks (perceptrons) face limitations

1980s: Connectionist revival, backpropagation

1990s: Statistical turn, field splits from AI

2000s: Kernel methods, ensemble learning

2012: ImageNet breakthrough, deep learning renaissance

The pattern: Incremental progress punctuated by breakthrough moments when scale, compute, and data aligned.